

# Product Requirements Document (PRD)

## JEMBY JAMES Electrical Digital Twin Platform

Next-Generation Physics-Aware Electrical Power Distribution Modeling, Field Survey Cockpit, & Single-Line Diagram Engine

| Document Info    | Details                                                                         |
|------------------|---------------------------------------------------------------------------------|
| Product Name     | JEMBY JAMES Electrical Digital Twin (JAMES)                                     |
| Document Version | 1.0.0                                                                           |
| Document Status  | Approved for Implementation                                                     |
| Target Audience  | Engineering Leadership, Field Surveyors, Electrical Engineers, Product Managers |
| Author           | JEMBY Solutions / Google Antigravity Architecture Team                          |
| Date             | September 2026                                                                  |

### 1. Executive Summary & Product Vision

#### 1.1 Mission Statement

The **JEMBY JAMES Electrical Digital Twin** is an intelligent, physics-grounded modeling platform that digitizes industrial and commercial electrical power distribution systems. It bridges the critical gap between field survey data collection, deterministic electrical engineering physics, multi-scale schematic visualization, and AI-assisted facility optimization.

#### 1.2 The Problem

Traditional electrical facility management and electrical study workflows suffer from systemic inefficiencies:

- 1. Error-Prone Manual Surveys:** Field engineers rely on paper clipboards, disconnected spreadsheets, and fragmented smartphone photos to capture complex panel schedules, switchgear nameplates, and feeder paths.
- 2. Static, Disconnected CAD Tools:** Legacy tools (AutoCAD, Visio, PDF one-lines) treat electrical drawings as dumb lines and text annotations. They possess zero understanding of electrical physics, upstream power flow, phase sequencing, or breaker ampacity constraints.
- 3. Rigid Database Schemas:** Electrical equipment exhibits hundreds of manufacturer-specific parameters (e.g. %Z for transformers, fuel controllers for generators, bypass-isolation for transfer switches). Rigid relational database schemas break whenever novel field equipment is surveyed.
- 4. High Latency from Survey to Single-Line Diagram (SLD):** Transforming field survey notes into validated single-line diagrams takes weeks of tedious drafting and back-and-forth verification.

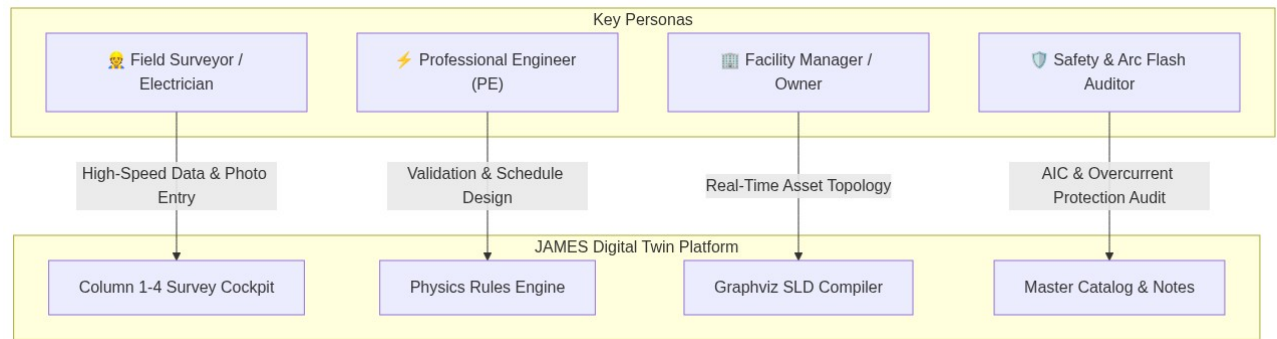
#### 1.3 The Solution

JAMES delivers a unified, high-speed **4-Column Field Collector Cockpit** and a **Deterministic Graph Engine**:

- High-Speed Touch-Optimized Surveying:** Rapid equipment instantiation with pre-populated engineering defaults across 10 equipment domains.
- Dual-Column Panel Schedule Matrix:** Interactive breaker-slot management with automatic phase calculation (A/B/C for 3-phase, A/B for 1-phase) and multi-pole ganged breaker row spans.

- **Contextual Dynamic Attributes (vendor\_attrs):** Context-aware nameplate inputs tailored to equipment archetypes, paired with an ad-hoc custom attribute slot for instant on-site adaptability.
- **Zero-Friction Photo & Note Capture:** Instant nameplate and observation photo capture (camera, gallery, drag-and-drop, clipboard paste) stored as self-contained payloads in SQLite and companion .jsonl.
- **Automated Multi-Scale SLD Generation:** Dynamic Graphviz DOT compiler that renders facility power flow graphs using cluster subgraphs, discrete rectangular port terminals, and polyline routing.

## 2. User Personas & Core Use Cases



### 2.1 Personas

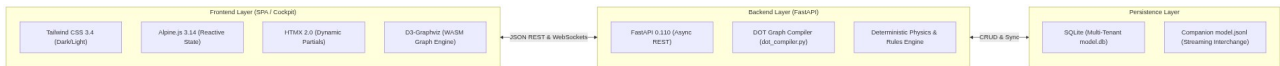
1. **Field Electrical Surveyor (Primary):** Needs to document complex electrical rooms in noisy, low-connectivity environments at maximum speed. Values keyboard shortcuts, big touch targets, auto-incrementing tags, and zero lost data. 2. **Professional Electrical Engineer / Designer:** Audits field data, validates transformer sizing, configures panel schedules, and exports single-line schematics for permits and studies. 3. **Facility Manager / Data Center Operator:** Uses the digital twin for real-time asset tracking, upstream power tracing (e.g., "Which breaker feeds Server Rack 12?"), and maintenance planning. 4. **Safety & Compliance Auditor (NFPA 70E / NEC):** Inspects equipment AIC ratings, conductor ampacity compliance, and working space clearances.

### 2.2 Primary User Flows

- **Use Case 1: Rapid Electrical Room Survey:** Surveyor creates a Main Distribution Panel (MDP-1), selects 480V 3Ø, populates 42 slots in minutes, attaches a nameplate photo, and defines upstream utility feeds.
- **Use Case 2: Multi-Scale Single-Line Visualization:** Engineer navigates from high-level facility power flow to drill down into a specific panel's internal breakers and downstream loads.
- **Use Case 3: Offline Data Export & Import:** System automatically persists to localized SQLite (model.db) and exports companion .jsonl for cloud ingestion and team sharing.

## 3. Product Architecture & Technical Specifications

JAMES implements a decoupled **Data-Physics-Visualization Triad**:



### 3.1 Technology Stack Table

| Component             | Technology             | Version       | Purpose                                                   |
|-----------------------|------------------------|---------------|-----------------------------------------------------------|
| Backend Framework     | Python / FastAPI       | 3.11+ / 0.110 | Asynchronous REST API, Graph Engine, Data Validation      |
| Data Validation       | Pydantic               | 2.6+          | Strongly-typed electrical twin data models                |
| Client Storage        | SQLite + JSONL         | SQLite 3      | Zero-config localized relational & line-delimited storage |
| Frontend Framework    | Alpine.js              | 3.14+         | Lightweight reactive state management                     |
| Styling Engine        | Tailwind CSS           | 3.4+          | Utility-first responsive design, dark mode, glassmorphism |
| Dynamic Interactivity | HTMX                   | 2.0+          | Server-driven UI updates and partial swapping             |
| Graph Visualization   | Graphviz + D3-Graphviz | WASM          | SVG rendering of single-line diagrams with pan/zoom       |

## 4. Functional Requirements (FR)

### 4.0 User Interface Architecture & Visual Layout

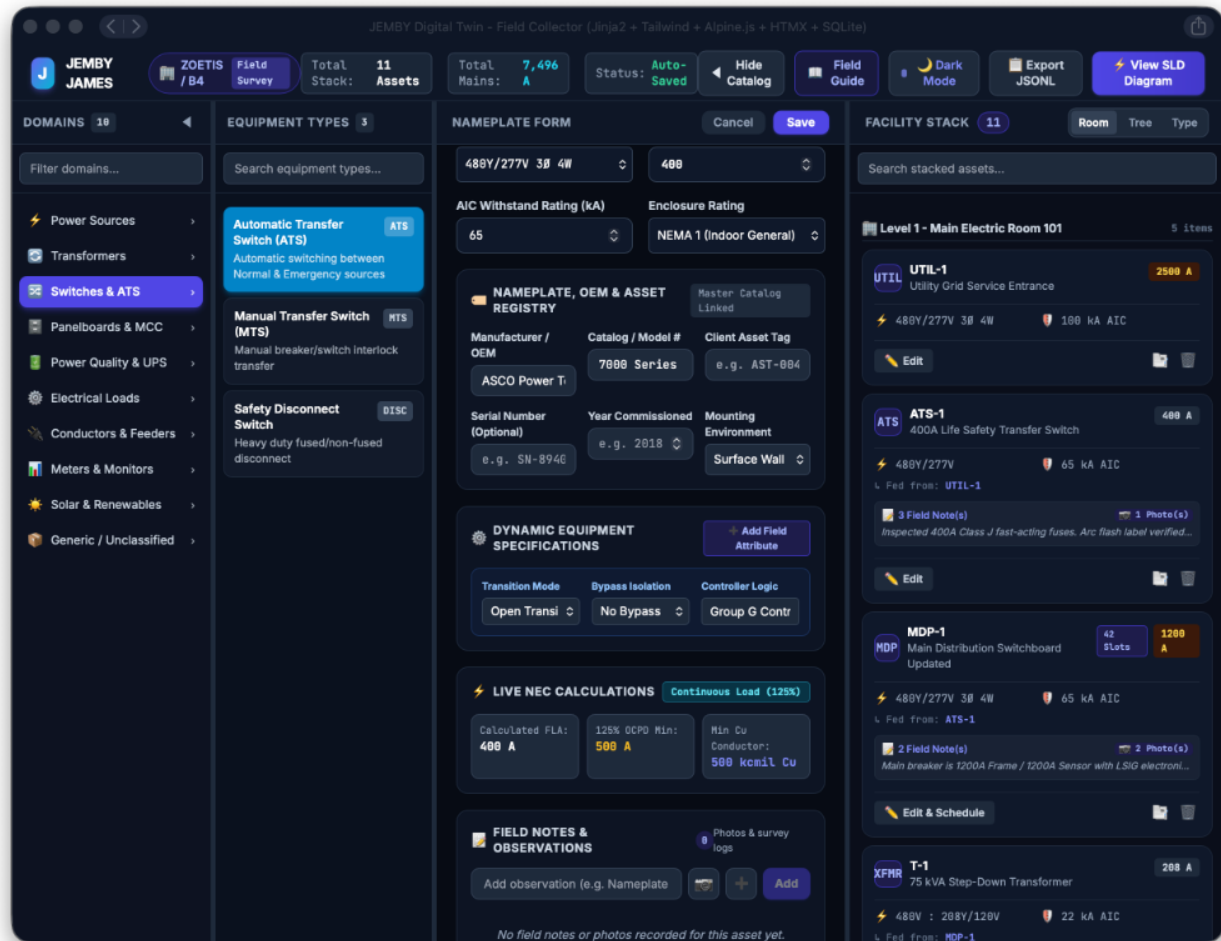


Figure 4.1: Production UI of the JEMBY JAMES 4-Column Field Collector Cockpit showing Domains (Col 1), Type Catalog (Col 2), Nameplate Form & Dynamic Specifications (Col 3), and Facility Stack with Grouping Modes (Col 4).

#### FR-1: 4-Column Resizable Survey Cockpit

##### • FR-1.1 Four-Column Layout:

- **Column 1 (Equipment Domains):** 10 categorized electrical domains (Power Sources, Transformers, Switches & ATS, Panelboards & MCC, Power Quality & UPS, Electrical Loads, Conductors & Feeders, Meters & Monitors, Solar & Renewables, Generic / Unclassified).
- **Column 2 (Type Catalog):** Archetype catalog with pre-configured ratings (e.g., LP, MDP, MCC, REC, GEN, XFMR, ATS, MTS, DISC, UPS, HVAC, MOTOR, BESS).
- **Column 3 (Form & Schedule Editor):** Physical ratings, Nameplate OEM registry, contextual dynamic slots, live NEC continuous load sizing, interactive dual-column panel schedule, and field observation notes.
- **Column 4 (Facility Stack & Flow Tree):** Grouped list of surveyed assets with real-time search filtering, status pills, and action controls.

##### • FR-1.2 Top Master Control Bar:

- **Facility Context:** Displays current multi-tenant client and facility identifiers (e.g., ZOETIS / B4 Field Survey).
- **Telemetry Metric Badges:** Real-time asset counter (Total Stack: N Assets), cumulative service entrance capacity (Total Mains: N A), and persistence health (Status: Auto-Saved).
- **Global Actions:**
  - Hide/Show Catalog toggle to expand the central editing canvas on laptop screens.
  - Field Guide modal trigger (? shortcut).
  - Dark Mode / Light Mode instant theme switch.
  - Export JSONL button to download streaming companion datasets.
  - ■ View SLD Diagram CTA button to instantly transition into the interactive graph visualizer.
- **FR-1.3 Stack Grouping Modes (Column 4):**
  - **Room Grouping (Room):** Groups equipment by physical location (e.g., ■ Level 1 - Main Electric Room 101, ■ Rooftop HVAC Penthouse).
  - **Upstream Tree Grouping (Tree):** Renders hierarchical power flow relationships from utility/generator feeds down to branch subpanels.
  - **Type Grouping (Type):** Categorizes assets by equipment classification.
- **FR-1.4 Splitter Resizing:** All column splitters shall support smooth, touch-friendly horizontal dragging with persisted width states.

## FR-2: Nameplate, OEM & Asset Registry

- **FR-2.1 OEM & Catalog Metadata:**
  - Manufacturer / OEM (with datalist autocomplete for major brands: *Square D / Schneider Electric, Eaton / Cutler-Hammer, Siemens, ABB / GE, ASCO Power Technologies, Caterpillar, Cummins, Kohler, Vertiv / Liebert, Olsun Electrics*)
  - Catalog / Model #
  - Client Asset Tag
  - Serial Number (Optional)
  - Year Commissioned
  - Mounting Environment (*Surface Wall Mount, Flush Recessed, Free-Standing Pad, Skid / Unit Substation*)
- **FR-2.2 Dedicated Nameplate Photo Slot:**
  - Integrated nameplate image upload supporting Camera, File Picker, Drag & Drop, and Clipboard Paste.
  - Displays thumbnail card with click-to-zoom Lightbox and deletion badge.

## FR-3: Dynamic Equipment Specifications & Ad-Hoc Attributes

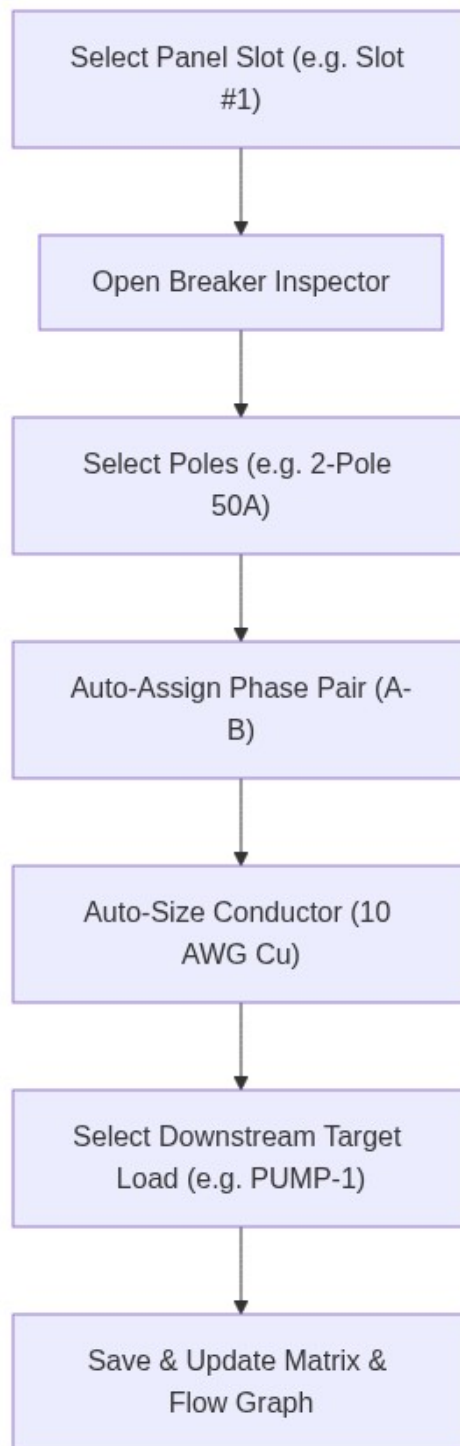
- **FR-3.1 Contextual Slots by Archetype:**
  - **Generators (GEN):** Standby/Prime kW rating, Fuel Type (*Diesel #2, Natural Gas, Propane, Dual Fuel*), Engine Controller (e.g., *EMCP 4.2B*).
  - **Transformers (XFMR / PAD):** Impedance (%Z), Temp Rise (*150°C, 115°C Low Loss, 80°C*), Cooling Class (*AA, FA, ONAN*).
  - **Transfer Switches (ATS / MTS):** Transition Mode (*Open Transition, Closed Transition / Soft-Load, Delayed Transition*), Bypass Isolation (*No Bypass, Dual Drawout Bypass*), Controller Logic (e.g., *Group G Controller*).
  - **Safety Disconnect Switches (DISC):** Fuse Class (*Class J, Class R, Class T, Class L, Class CC, Non-Fused*), Installed Fuse Amps, Blown Fuse Indicators (Yes/No).
- **FR-3.2 Ad-Hoc Nameplate Extensibility (■ Add Field Attribute):**
  - Provides on-the-fly creation of arbitrary Parameter Name : Value pairs for rare field equipment.
  - Scoped to that asset instance and saved inside `attributes.vendor_attrs` without schema migrations.

#### FR-4: Real-Time Live NEC Calculations

- **FR-4.1 Continuous Load Sizing (NEC 125% Rule):**
  - Automatically calculates Full Load Amps (Calculated FLA).
  - Computes minimum Overcurrent Protection Device rating:  $OCPD_{min} = 1.25 \times FLA$ .
  - Auto-recommends minimum copper feeder conductor gauge (e.g., 125% OCPD Min: 500 A → Min Cu Conductor: 500 kcmil Cu).

#### FR-5: Dual-Column Panel Schedule Matrix

- **FR-5.1 Physical Slot Configuration:**
  - Supports standard panel sizes (12, 18, 24, 30, 42, 54, 72, 84 slots).
  - Auto-calculates electrical phase:
  - 3-Phase Panels: Cycle sequence  $A \rightarrow B \rightarrow C \rightarrow A \dots$  (Phases color-coded: A=Red, B=Blue, C=Emerald).
  - 1-Phase Panels: Alternating  $A \rightarrow B \rightarrow A \dots$
- **FR-5.2 Breaker Inspector Modal:**
  - Clicking any slot opens the inspector modal to configure: Circuit Description, Trip Amps (15A–1200A), Poles (1P, 2P, 3P), Type (*MCCB*, *Miniature*, *AFCI*, *GFCI*, *MCP*, *Drawout*), Wire Gauge (auto-computed from ampacity), and Downstream Target Load.
  - **Multi-Pole Ganging:** Setting 2-pole or 3-pole automatically locks and groups sequential phase rows with multi-row span indicators.
  - **One-Click Space Conversion:** Remove Breaker button clears slot to Space/Spare.



#### FR-6: Field Notes, Observations & Photo Attachments

- **FR-6.1 Multi-Input Capture:** Camera snaps, device gallery multi-upload, drag & drop, and clipboard paste (Cmd+v / Ctrl+v).
- **FR-6.2 Auto-Flush Safety:** If a surveyor attaches photos or types an observation and immediately hits "**■ Save Asset**" without clicking "Add", the system automatically flushes the staged content directly into the asset's permanent notes.
- **FR-6.3 Fullscreen Lightbox Modal:** High-resolution zoomable viewer with photo timestamp, caption, and download button.

## FR-7: Multi-Scale Graphviz Single-Line Diagram Engine

- **FR-7.1 10-Domain Harmonized Pastel Palette:**

- Equipment clusters are color-coded by electrical domain with high-contrast pastel background fills, distinct border outlines, and dark slate typography (#0F172A):
- **Power Sources** (sources): Amber (#FEF3C7 / #D97706)
- **Transformers** (transformers): Purple (#F3E8FF / #9333EA)
- **Switches & ATS** (switches): Royal Blue (#DBEAFE / #2563EB)
- **Panelboards & MCC** (panels): Slate / Indigo (#F1F5F9 / #475569)
- **Power Quality & UPS** (power\_quality): Emerald (#D1FAE5 / #059669)
- **Loads & Chillers** (loads): Teal (#CCFBF1 / #0D9488)
- **Feeders & Busway** (cables): Orange (#FFEDD5 / #EA580C)
- **Meters & Monitors** (metering): Sky (#E0F2FE / #0284C7)
- **Solar & Renewables** (renewables): Yellow (#FEF9C3 / #CA8A04)
- **Generic / Assets** (generic): Slate (#F8FAFC / #64748B)

- **FR-7.2 Silver Blueprint Drafting Canvas:**

- The Single-Line Diagram viewport renders over an architectural silver metallic drafting canvas (#CBD5E1) featuring a subtle dot grid matrix for maximum line contrast and drawing clarity.

- **FR-7.3 Discrete Port Terminals & Dedicated Slot Numbering:**

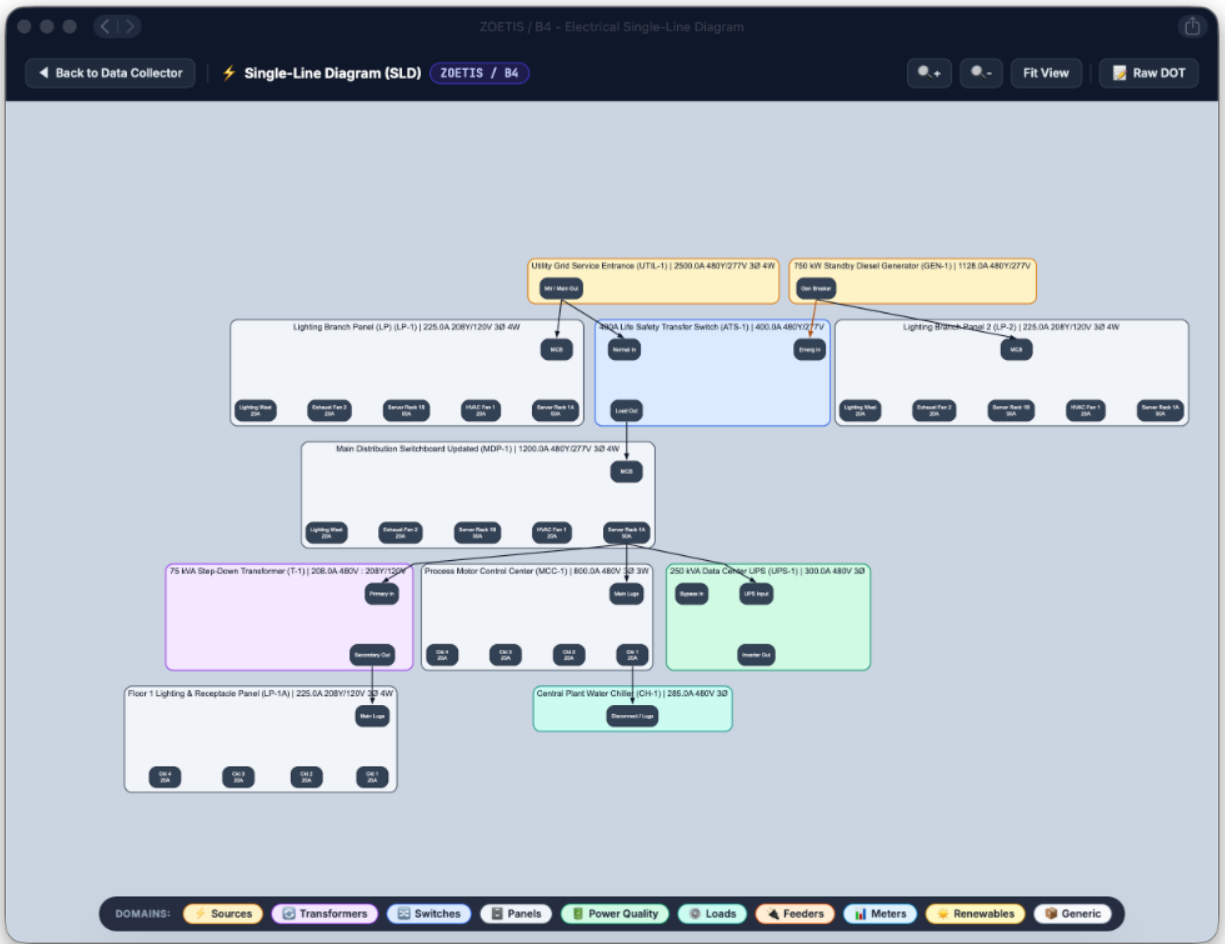
- Connections terminate at physical rectangular ports (fillcolor="#334155", fontcolor="#FFFFFF").
- Circuit breaker and feeder slots feature a dedicated top-row physical slot tag:
- **Line 1:** [ Slot 1 ] (*Physical Slot / Position ID*)
- **Line 2:** Server Rack 1A (*Circuit / Load Description*)
- **Line 3:** 50A (*Breaker Trip Rating*)

- **FR-7.4 Orthogonal Polyline Routing & Dark Feeder Lines:**

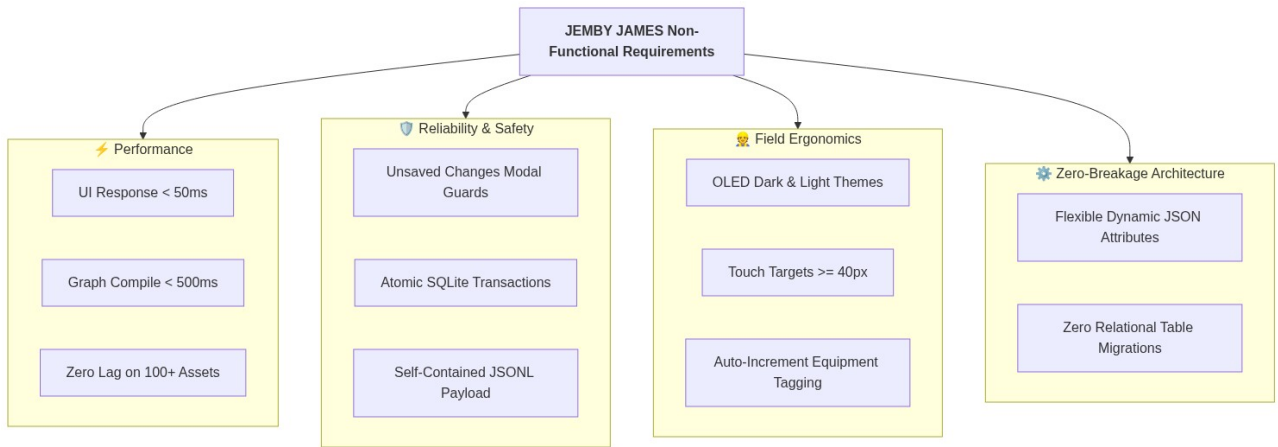
- Feeder runs are routed with splines=polyline in bold dark slate (#0F172A, penwidth=2.0), with emergency standby generator feeds highlighted in safety amber (#B45309, penwidth=2.2).

- **FR-7.5 Floating Interactive Domain Legend:**

- Persistent pill bar docked at the viewport base providing instant domain context.



## 5. Non-Functional Requirements (NFR)



### 5.1 Performance & Responsiveness

- **NFR-1.1:** The Cockpit UI shall respond to user input in under **50 milliseconds**.

- **NFR-1.2:** Full facility DOT graph compilation and SVG rendering shall complete in under **500 milliseconds** for facilities with up to 250 assets.
- **NFR-1.3:** The client application bundle shall operate with zero external npm build-step dependencies when running standalone.

## 5.2 Reliability & Data Integrity

- **NFR-2.1: Unsaved Changes Safety Guard:** Any attempt to switch equipment domains, select a new type, or navigate away while uncommitted edits or staged photos exist shall trigger an explicit confirmation modal (*Save & Switch*, *Discard*, *Keep Editing*).
- **NFR-2.2 Unique Tag Enforcement:** The system shall validate equipment tags in real-time and prevent duplicate tags within the same facility.
- **NFR-2.3 Electrical Cycle Prevention:** The upstream Fed From selector shall automatically filter out self, downstream descendants, and incompatible source domains to prevent physical power flow loops.

## 5.3 Usability & Field Ergonomics

- **NFR-3.1:** Support high-contrast **OLED Dark Mode** (for dimly lit electrical rooms) and **Clean Light Mode** (for bright outdoor substations), with automatic persistence.
  - **NFR-3.2:** Minimum interactive touch target size of **40 × 40 px** across all mobile and tablet inputs.
  - **NFR-3.3:** Keyboard shortcuts: ? toggles Field Survey Help Modal, Enter saves/adds notes and breakers, Esc dismisses modals.
-

## 6. Data Model & Schema Specification

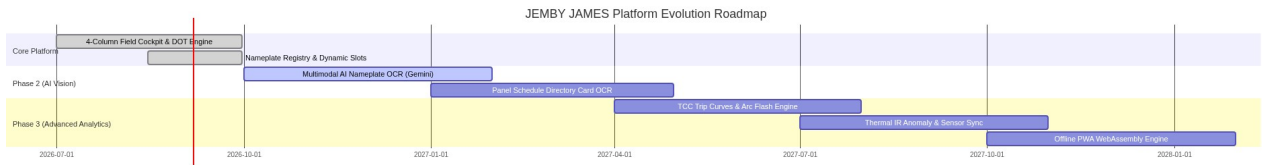
### 6.1 Unified Node Schema (DigitalTwinNode)

```
{
  "id": "s_1788691000",
  "tag": "MDP-1",
  "name": "Main Distribution Panel",
  "domain": "panels",
  "type_tag": "MDP",
  "type_name": "Main Distribution Panel (MDP)",
  "room": "Main Electrical Room 101",
  "fed_from": "ATS-1",
  "voltage": "480Y/277V 3Ø 4W",
  "amps": 1200.0,
  "aic": 65.0,
  "is_panel": 1,
  "slots": 42,
  "survey_sequence": 1,
  "attributes": {
    "manufacturer": "Square D / Schneider Electric",
    "model_no": "QED-2",
    "asset_tag": "AST-ZOETIS-0042",
    "serial_no": "SN-894021-B",
    "year_installed": 2018,
    "mounting": "Freestanding",
    "main_type": "MCB",
    "nameplate_photo": {
      "id": "photo_1788691001",
      "name": "MDP1_Nameplate.jpg",
      "dataUrl": "data:image/jpeg;base64,...",
      "timestamp": "10:45 AM"
    }
  },
  "vendor_attrs": {
    "section_type": "Main + Distribution",
    "main_device": "PowerPact P-Frame",
    "bus_bracing_ka": 65
  },
  "schedule": [
    {
      "leftSlot": 1,
      "leftDesc": "Server Rack 1A",
      "leftTrip": "50",
      "leftPoles": 2,
      "leftType": "MCCB",
      "leftWire": "10 AWG Cu",
      "leftTargetLoad": "PDU-1",
      "phase": "A",
      "phaseColor": "text-red-500",
      "rightSlot": 2,
      "rightDesc": "HVAC Fan 1",
      "rightTrip": "20",
      "rightPoles": 1,
      "rightType": "MCCB",
      "rightWire": "12 AWG Cu",
      "rightTargetLoad": "CH-1"
    }
  ],
  "notes": [
    {
      "id": "note_1788691002",
      "text": "Infrared scan completed. Lugs torqued to 275 in-lbs.",
      "date": "09/06 10:45",
      "author": "Field Survey",
      "photos": []
    }
  ]
}
```

7. REST API Endpoints

| Method | Endpoint                                               | Description                                                    |
|--------|--------------------------------------------------------|----------------------------------------------------------------|
| GET    | /cockpit                                               | Serves the interactive 4-Column Field Collector Cockpit        |
| GET    | /sld                                                   | Serves the interactive Single-Line Diagram viewer              |
| GET    | /api/clients/{client}/facilities/{facility}/nodes      | Retrieves all equipment nodes for a facility                   |
| POST   | /api/clients/{client}/facilities/{facility}/nodes      | Upserts an asset node (with schedule, photos, & dynamic attrs) |
| DELETE | /api/clients/{client}/facilities/{facility}/nodes/{id} | Deletes an asset node and cleans up downstream references      |
| GET    | /api/clients/{client}/facilities/{facility}/dot        | Compiles facility digital twin graph to Graphviz DOT string    |
| GET    | /api/catalog                                           | Retrieves master equipment domains and type definitions        |

8. Strategic Roadmap & AI Futures



- **Phase 2.1: Multimodal AI Nameplate OCR** (Gemini 2.5 Flash agent extracts voltage, amps, AIC, model, and serial directly from photos).
- **Phase 2.2: Panel Directory OCR** (Digitizes paper directory cards into dual-column schedule matrix automatically).
- **Phase 3.1: Parametric TCC Curves & Selective Coordination** (Mathematical breaker trip curves and arc-flash clearing time estimation).
- **Phase 3.2: Thermal IR Integration** (Attaching FLIR radiometric images to breaker slots to flag hot spots).
- **Phase 3.3: Offline PWA** (Client-side WASM SQLite sync for zero-connectivity environments).

9. Success Metrics & Key Performance Indicators (KPIs)

1. **Survey Velocity:** Reduce total on-site data capture and nameplate logging time by > **65%**. 2. **SLD Turnaround Time:** Accelerate draft Single-Line Diagram generation from **2–3 weeks to instant real-time compilation**. 3. **Data Completeness:** Achieve **100% asset nameplate capture** including photos and dynamic vendor parameters. 4. **Schema Stability:** **Zero database schema migrations** required when adding new equipment types or vendor attributes. 5. **Test Coverage:** Maintain > **95% test coverage** across all core physics rules, DOT compilers, and API routes.